

Quiz 2: Adversarial and Sequential Decision Making

Weeks 3-5

CMPSC 480 - Student Question Deck

Assessment overview

Trace adversarial backups, solve a Bellman step, and apply a Q-learning update.

Coverage: Weeks 3-5

Points

20 total

Expected time

35 minutes

Submission

Individual PDF

Learning objectives

- ▶ Trace minimax and pruning.
- ▶ Interpret an MDP.
- ▶ Perform Bellman and Q-learning updates.
- ▶ Distinguish planning from learning.

Requirements checklist

- ▶ Work independently.
- ▶ Show backed-up values and arithmetic.
- ▶ State terminal conventions.
- ▶ No electronic communication.

Task 1 - Minimax and alpha-beta (6 points)

A MAX root has MIN children A and B. A has leaves 3 and 5; B has leaves 2 and 9.

- a. Compute the minimax value and selected branch. [2 points]
- b. Search A first. After B's first leaf 2, can B's second leaf be pruned? [2 points]
- c. Does pruning change the backed-up value? Explain. [2 points]

Task 2 - MDP backup (6 points)

From state s , action a reaches terminal $+5$ with probability 0.8 and state t with probability 0.2 . The immediate reward is 0 , $\gamma=0.9$, and $V(t)=2$.

- a. Write the Bellman action-value expression. [2 points]
- b. Compute $Q(s,a)$. [2 points]
- c. Explain what value iteration maximizes. [2 points]

Task 3 - Q-learning (4 points)

$Q(s,a)=1$, reward=2, max next $Q=3$, gamma=0.5, and alpha=0.25.

a. Compute the nonterminal target and TD error.

[2 points]

b. Compute the updated Q value and state the terminal change.

[2 points]

Task 4 - Planning versus learning (4 points)

Compare value iteration and Q-learning for the same gridworld.

- a. What information does each method require? [2 points]
- b. Why must a Q-learning evaluation run use epsilon zero? [2 points]

Edge cases and defensive programming

- ▶ A minimax root is terminal.
- ▶ An MDP action has probabilities that do not sum to one.
- ▶ A Q-learning transition is terminal.

Submission instructions

- ▶ Show every backup and update.
- ▶ State any convention needed for terminal reward.
- ▶ Submit only the completed response PDF.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric

Category	Points	Required evidence
Minimax and alpha-beta	6	Tree trace.
MDP backup	6	Expected-return calculation.
Q-learning	4	One numerical update.
Planning versus learning	4	Short explanation.

Reflection prompt

In one sentence:

State one reason a learned policy can differ from a planning oracle.